

Alexander J Root

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EDUCATION

Stanford University

Ph.D. in Computer Science

• Advisor: Fredrik Kjolstad

09/2022 - Present

GPA: 4.0/4.0

Massachusetts Institute of Technology

M.Eng. in Electrical Engineering & Computer Science

• Advisors: Jonathan Ragan-Kelley and Andrew Adams

• Thesis: Optimizing Vector Instruction Selection for Digital Signal Processing

06/2021 - 06/2022

GPA: 5.0/5.0

Massachusetts Institute of Technology

S.B. in Computer Science & Engineering

• Advisors: Fr  do Durand and Jonathan Ragan-Kelley

• Bachelor's Project: High Performance Image Processing with Fixed-Point Types

09/2017 - 06/2021

GPA: 5.0/5.0

RESEARCH EXPERIENCE

Stanford Compilers Group

PhD Student Researcher under Fredrik Kjolstad

• Researching domain-specific compilation techniques for high-performance irregular computations. I have developed compiler techniques for both sparse array programming [3, 4, 5, 8] and spatial computing (e.g. ray-tracing and collision detection) [1, 2].

Sept 2022 - Present

Stanford, USA

MIT Visual Computing Languages & Systems Group

Undergraduate and Master's Researcher under Jonathan Ragan-Kelley, Andrew Adams, and Fr  do Durand.

• Researched projects related to high-performance digital signal processing, including automatic quantization, bounds inference, and vector instruction selection [6, 7].

May 2019 - Aug 2022

Cambridge, USA

PUBLICATIONS

Refereed Conference and Journal Papers

- [1] Christophe Gyurgyik, **Alexander J Root**, Fredrik Kjolstad, "Decoupling Data Layouts from Bounding Volume Hierarchies". In: *Proceedings of the ACM on Programming Languages* 10. **PLDI** (June 2026). DOI: [10.1145/3808253](#). URL: [\[link\]](#)
- [2] **Alexander J Root**, Christophe Gyurgyik, Purvi Goel, Kayvon Fatahalian, Jonathan Ragan-Kelley, Andrew Adams, Fredrik Kjolstad, "Bonsai: Compiling Queries to Pruned Tree Traversals". In: *Proceedings of the ACM on Programming Languages* 10. **PLDI** (June 2026). **Distinguished Paper Award**. DOI: [10.1145/3808256](#). URL: [\[link\]](#)
- [3] Bobby Yan, **Alexander J Root**, Trevor Gale, David Broman, Fredrik Kjolstad, "Fast Autoscheduling for Sparse ML Frameworks". In: *Proceedings of the 24th ACM/IEEE International Symposium on Code Generation and Optimization*. **CGO '26**. Sydney, Australia: Association for Computing Machinery, 2026. DOI: [10.1109/CGO68049.2026.11394842](#).
- [4] **Alexander J Root**, Bobby Yan, Peiming Liu, Christophe Gyurgyik, Aart J.C. Bik, Fredrik Kjolstad, "Compilation of Shape Operators on Sparse Arrays". In: *Proc. ACM Program. Lang.* 8. **OOPSLA2** (Oct. 2024). DOI: [10.1145/3689752](#). URL: [\[link\]](#)
- [5] Peiming Liu, **Alexander J Root**, Anlun Xu, Yinying Li, Fredrik Kjolstad, Aart J.C. Bik, "Compiler Support for Sparse Tensor Convolutions". In: *Proc. ACM Program. Lang.* 8. **OOPSLA2** (Oct. 2024). DOI: [10.1145/3689721](#). URL: [\[link\]](#)
- [6] **Alexander J Root**, Maaz Bin Safeer Ahmad, Dillon Sharlet, Andrew Adams, Shoaib Kamil, Jonathan Ragan-Kelley, "Fast Instruction Selection for Fast Digital Signal Processing". In: *Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 4*. **ASPLOS '23**. Vancouver, BC, Canada: Association for Computing Machinery, 2024, pp. 125–137. ISBN: 9798400703942. DOI: [10.1145/3623278.3624768](#). URL: [\[link\]](#)
- [7] Maaz Bin Safeer Ahmad, **Alexander J Root**, Andrew Adams, Shoaib Kamil, Alvin Cheung, "Vector instruction selection for digital signal processors using program synthesis". In: *Proceedings of the 27th ACM International Conference on Architectural Support for Programming Languages and Operating Systems*. **ASPLOS '22**. Lausanne, Switzerland: Association for Computing Machinery, 2022, pp. 1004–1016. ISBN: 9781450392051. DOI: [10.1145/3503222.3507714](#). URL: [\[link\]](#)

Under Review

- [8] Atharva Chougule*, **Alexander J Root***, Rubens Lacouture, Bobby Yan, Rohan Yadav, Fredrik Kjolstad, *Partitioning Unstructured Sparse Tensor Algebra for Load-Balanced Parallel Execution*. Under Review. 2026. Note: * denotes equal contribution.

WORK EXPERIENCE

- NVIDIA Research** May 2025 - Present
Research Intern (Real-Time Graphics) with Matt Pharr and Rohan Sawhney Santa Clara, USA
• Designing a scheduling language for parallel divergent workloads (e.g. path tracing or WoS).
- Adobe Research** May 2023 - Aug 2023
Research Intern (Compilers) with Andrew Adams San Francisco, USA
• Designed a DSL for geometric queries that decouples algorithm from accelerator structures [2, 1].
- Adobe Research** June 2022 - Nov 2022
Research Intern (Compilers) with Maaz Bin Safeer Ahmad, Andrew Adams, and Shoaib Kamil Virtual, USA
• Developed a language and system to improve fixed-point vector instruction selection for the Halide compiler [6].
- Adobe Research** June 2021 - Dec 2021
Research Intern (Compilers) with Andrew Adams and Shoaib Kamil Virtual, USA
• Developed techniques for constant bounds approximations and vector instruction selection [7].
- Intel** Jan 2021 - May 2021
Research Intern (Compilers) with Derek Gerstmann Virtual, USA
• Designed a new autoscheduler for Halide. Achieved $2\times$ faster compilation with $1.06\times$ faster runtime.
- Microsoft** June 2020 - Sept 2020
Software Engineering Intern Virtual, USA
• Contributed to verification infrastructure for access of control of virtual machines.
- Lawrence Livermore National Lab** June 2019 - Sept 2019
Computation Intern Livermore, USA
• Developed distributed numerical optimization methods in C++.
- Iterative Scopes (now Iterative Health)** Feb 2018 - Aug 2018
Associate Software Engineer Cambridge, USA
• Automated and tested large scale image processing and machine vision systems.
- Redding Electric Utility** June 2017 - Aug 2017
Engineering Intern Redding, USA
• Implemented query and reporting systems in C++ for financial data sets.

TEACHING EXPERIENCE

- CS 343S: Domain-Specific Language Design Studio** Spring 2024 and 2025
Teaching Fellow/Lecturer Stanford University
• (co-)Developed and taught new class on design and implementation of domain-specific languages.
- CS 343D: Domain-Specific Programming Models and Compilers** Winter 2023 and 2024
Course Assistant under Fredrik Kjolstad Stanford University
• Developed coursework, gave lectures, ran office hours, graded assignments, and managed course logistics for a research seminar on domain-specific languages.
- 6.818: Dynamic Computer Language Engineering** Fall 2021
Teaching Assistant under Michael Carbin MIT
• Gave lectures, ran office hours, graded assignments, and managed course logistics for a project-oriented class on designing JIT compilers.
- 6.006: Introduction to Algorithms** Spring 2019 and 2020
Teaching Assistant under Jason Ku MIT
• Taught 20 student recitations twice a week, ran office hours, and graded assignments for introductory algorithms.

MENTORING

- Atharva Chougule** Fall 2025 - Present
Stanford M.S. (expected) 2027 Stanford
• Investigating methods for compiling sparse tensor algebra to load-balanced parallel kernels [8].
- Devanshu Ladsaria** Summer 2024 - Present
Stanford B.S. 2025 and M.S. (expected) 2026 Stanford
• Developing a DSL for specifying complex layouts for path tracing queues, building on the ideas of SCION.
- Christophe Gyurgyik** Summer 2024 - Winter 2026
Stanford Ph.D. (expected) 2028 Stanford
• Chris and I developed SCION [1] as an extension of our work on BONSAI [2]. We show that separating data structure semantics from memory layout optimizations provides useful design space exploration for tree queries.

Ishita Gupta

Stanford B.S. 2025 and M.S. (expected) 2026

Fall 2024 - Spring 2025

Stanford

- Explored designing a domain-specific memory allocator for high-performance CPU renderers.

Katherine Mohr

MIT S.B. 2023 and M.Eng. 2024, now Stanford CS Ph.D. student

2021-2022

MIT

- Developed a compiler for pattern matching rewrite rules in the Halide compiler, to improve the performance of term-rewriting-systems (extensively used throughout code generation).

Mario Leyva

MIT S.B. 2022 and M.Eng. 2023, now Software Engineer at Google

Spring 2022

MIT

- Implemented fixed-point Porter-Duff image compositing algorithms in Halide.

Evan Lee

University of Waterloo, B.S. 2024, now Compiler Engineer at Meta

Summer 2021

Google Summer of Code

- The first Google Summer of Code intern for the Halide language. Evan ran an [evaluation](#) of the performance impact of adding synthesized rewrite rules to the Halide expression simplifier, showing that only 10% of the rules were necessary to achieve the performance gains promised in the paper.

HONORS AND AWARDS

NVIDIA Graduate Fellowship	2026 - 2027
Distinguished Paper Award (PLDI) for BONSAI [2]	2026
Qualcomm Innovation Fellowship	2025 - 2026
National Science Foundation Graduate Research Fellowship (NSF GRFP)	2022 - 2025
School of Engineering Fellowship, Stanford	2022 - 2023
National Engineering Honor Society Member, Tau Beta Pi (TBP)	2020 - 2021
National Electrical Engineering Honor Society Member, IEEE-Eta Kappa Nu (HKN)	2019 - 2021
Keel Foundation Undergraduate Research and Innovation Scholar, MIT	2019 - 2020

INVITED TALKS

Scheduling Divergent and Data-Dependent Recursive Workloads for Parallel Architectures Stanford CS 343D Guest Lecture	Stanford, CA Mar 2026
Decoupling Spatial Queries from Accelerator Trees Stanford SystemX Alliance	Stanford, CA Nov 2025
Decoupling Spatial Queries from Accelerator Trees Qualcomm Codegen Seminar	Virtual Oct 2025
Decoupling Spatial Queries from Accelerator Trees PORTAL Retreat	Half Moon Bay, CA Aug 2025
Decoupling Spatial Queries from Accelerator Trees Stanford CS 343D Guest Lecture	Stanford, CA Feb 2025
Decoupling Spatial Queries from Accelerator Trees Samsung Invited Talk	Virtual June 2024
Fast Instruction Selection for Fast Digital Signal Processing UCSD Graphics Seminar	La Jolla, CA Apr 2023

SERVICE

SIGGRAPH External Reviewer	2026
Eurographics External Reviewer	2026
Transactions on Graphics (ToG) External Reviewer	2025
SIGGRAPH Asia External Reviewer	2023 - 2026